

Sparse Forecast Combination for a Single Asset

Problem Setting

We aim to combine multiple forecasting models to improve the accuracy of predictions for a single asset (e.g., price, demand, or return). Our objective is to minimize forecast risk while selecting only a small number k of models.

We learn non-negative, sparse, and normalized weights $\mathbf{w}$ over the forecasting models, subject to the following constraints:

- $\sum_i w_i = 1$ (weights sum to one),
- $w_i \geq 0$ (non-negative weights),
- $\|\mathbf{w}\|_0 \leq k$ (at most k non-zero weights).

Notation

- n : number of forecasting models
- T : number of time steps
- y_t : true value of the asset at time t
- $\hat{y}_{i,t}$: prediction of model i at time t
- $e_{i,t} = \hat{y}_{i,t} - y_t$: forecast error

Let the error matrix be:

$$E = \begin{bmatrix} e_{1,1} & \cdots & e_{1,T} \\ \vdots & \ddots & \vdots \\ e_{n,1} & \cdots & e_{n,T} \end{bmatrix} \in \mathbb{R}^{n \times T}$$

Optimization Objective

We seek weights $\mathbf{w} \in \mathbb{R}^n$ that combine forecasts:

$$\hat{y}_t^{\text{combined}} = \sum_{i=1}^n w_i \hat{y}_{i,t}$$

We minimize the forecast risk:

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top A \mathbf{w} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq 0, \quad \|\mathbf{w}\|_0 \leq k \quad (1)$$

where A is the forecast risk matrix. Two options for defining A :

- Empirical covariance of forecast errors:

$$A = \frac{1}{T} E E^\top$$

- Time-decayed covariance:

$$A = \sum_{t=1}^T \lambda^{T-t} e_t e_t^\top \quad \text{for } \lambda \in (0, 1)$$

Interpretation

This approach is analogous to mean-variance optimization in portfolio selection but applied to forecast combination. The goal is to:

- Reduce forecast error (diagonal of A),
- Reduce redundancy (off-diagonal of A),
- Achieve interpretability via sparsity.

Use Case

Applicable to forecasting problems in:

- Electricity load
- Stock prices
- Weather prediction

1 How to Solve Problem (1)

Using the algorithm I have provided in my paper with George.

Extensions

- Relax ℓ_0 with ℓ_1 penalty:

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^\top A \mathbf{w} + \mu \|\mathbf{w}\|_1 \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \mathbf{w} \geq 0$$

- Use rolling windows or exponential decay to update A over time.
- Consider incorporating expected return of each model (if defined).

Problem Setup

Suppose we are forecasting returns for m assets, indexed by $j = 1, \dots, m$. For each asset j , we have predictions from n different forecasting models. Let $w^{(j)} \in \mathbb{R}^n$ represent the aggregation weights over the n predictors for asset j , constrained to be nonnegative and sum to one:

$$\sum_{i=1}^n w_i^{(j)} = 1, \quad w_i^{(j)} \geq 0 \quad \text{and} \quad \|w^{(j)}\|_0 \leq k.$$

We assume a risk (e.g., mean-squared forecast error or realized return variance) associated with each asset's weighted forecast is quantified by a symmetric positive semidefinite matrix $A^{(j)} \in \mathbb{R}^{n \times n}$, derived from rolling covariance of forecast errors for asset j .

Multi-Asset Aggregation Objective (Summation Form)

We aim to minimize the **aggregate forecast risk across all assets**:

$$\min_{w^{(1)}, \dots, w^{(m)}} \quad \frac{1}{2} \sum_{j=1}^m (w^{(j)})^\top A^{(j)} w^{(j)} \quad \text{subject to} \quad \begin{cases} \mathbf{1}^\top w^{(j)} = 1, \\ w^{(j)} \geq 0, \\ \|w^{(j)}\|_0 \leq k, \end{cases} \quad \text{for } j = 1, \dots, m.$$

Matrix Reformulation Using $W \in \mathbb{R}^{n \times m}$

Define:

$$W = \begin{bmatrix} | & | & & | \\ w^{(1)} & w^{(2)} & \dots & w^{(m)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times m}$$

Then the objective function becomes:

$$\frac{1}{2} \sum_{j=1}^m (w^{(j)})^\top A^{(j)} w^{(j)} = \frac{1}{2} \sum_{j=1}^m \text{tr} \left((w^{(j)})^\top A^{(j)} w^{(j)} \right) = \frac{1}{2} \sum_{j=1}^m \text{tr} \left(A^{(j)} w^{(j)} (w^{(j)})^\top \right).$$

Special Case: Homogeneous Risk Structure

If $A^{(j)} = A$ for all j , the expression simplifies to:

$$\frac{1}{2} \sum_{j=1}^m (w^{(j)})^\top A w^{(j)} = \frac{1}{2} \text{tr} \left(W^\top A W \right).$$

This expression is valid because: - $W^\top A W \in \mathbb{R}^{m \times m}$, - and $\text{tr}(W^\top A W) = \sum_{j=1}^m (w^{(j)})^\top A w^{(j)}$ due to trace cyclicity and summing over diagonal.

Final Joint Optimization Problem

$$\min_{W \in \mathbb{R}^{n \times m}} \quad \frac{1}{2} \text{tr}(W^\top A W) \quad \text{subject to} \quad \begin{cases} W^\top \mathbf{1}_n = \mathbf{1}_m, \\ W \geq 0, \\ \|w^{(j)}\|_0 \leq k \quad \text{for } j = 1, \dots, m. \end{cases}$$

Note: The sparsity constraint is enforced column-wise.